

超越QSAR：用于增强药物毒性预测的定量知识–活性关系（QKARs）

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摘要

计算毒理学在风险评估和药物安全方面起着重要作用。该领域传统上由定量构效关系（QSARs）主导，其基于化学结构单独预测毒理效应。尽管QSARs取得了一些成功，但其结构依赖性限制了药物毒性预测，因为细微的结构改动可能导致显著的毒性变化。人工智能（AI）的进步，尤其是文本嵌入和生成式AI，为通过利用更广泛的化学知识及其与结构数据的整合来增强毒性预测提供了机会。在本研究中，我们提出了一种新颖的框架——定量知识-活性关系（QKARs），该框架使用特定领域知识预测毒性。我们利用三种不同的知识表示开发了QKAR模型，用于两个药物毒性终点，即药物性肝损伤（DILI）和药物性心脏毒性（DICT）。

关键词：计算毒理学；定量知识–活性关系（QKARs）；药物诱导的心脏毒性；药物诱导的肝损伤；定量构效关系（QSARs）；新方法学（NAMs）

计算毒理学侧重于使用计算机模拟方法预测化学物质对生物体的毒性影响；它在毒理学研究和药物安全性研究中发挥着日益重要的作用（Kavlock 等，2008；Muster 等，2008；Li 等，2021, 2023；Sinha 等，2023；Haßmann 等，2024；Li 等，2024；Qu 等，2025），特别是在人工智能（AI）和机器学习（ML）进展的推动下。监管机构也鼓励使用计算方法来支持安全性评估。例如，2025 年 4 月，美国食品药品监督管理局（FDA）发布了《FDA 减少临床前安全性研究动物测试路线图》https://www.fda.gov/files/newsroom/published/roadmap_to_reducing_animal_testing_in_preclinical_safety_studies.pdf

mal_testing_in_preclinical_safety_studies.pdf，强调在消费品安全评估中采用新方法学（NAMs），包括 AI 模型。该路线图还得到了 FDA 现代化法案 3.0 的支持。

<https://www.congress.gov/bill/119th-congress/senate-bill/355>，表明 FDA 在现代化毒理学评估和减少对动物使用依赖方面的更广泛承诺。

自计算毒理学诞生以来，它一直由定量构效关系（QSAR）方法主导，这些方法基于化学结构决定其生物学或毒理学活性的原理（Tropsha 等，2024）。因此，QSAR 模型广泛采用机器学习（ML）根据由化学描述符表示的化学结构预测毒性（Popova 等，2018；Muratov 等，2020；Li 等，2021；Pérez Santín 等，2021；Chen 等，2023；Moret 等，2023）。尽管 QSAR 在毒理学应用中表现出相当大的成功（Nantasenamat 等，2010），但其完全依赖化学结构的特点限制了其应用范围。这一缺点在药物毒性评估中尤为突出，因为微小的结构修改可能

Beyond QSARs: Quantitative Knowledge–Activity Relationships (QKARs) for enhanced drug toxicity prediction

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Abstract

Computational toxicology plays an important role in risk assessment and drug safety. The field has been traditionally dominated by Quantitative Structure–Activity Relationships (QSARs), which predict toxicological effects based solely on chemical structure. Although QSARs have achieved successes, their structure reliance limits drug toxicity predictions, where small structural modifications may cause major toxicity changes. Advances in artificial intelligence (AI), especially text embedding and generative AI, provide an opportunity to enhance toxicity predictions by leveraging broader chemical knowledge and its integration with structural data. In this study, we propose a novel framework, Quantitative Knowledge–Activity Relationships (QKARs), which predicts toxicity using domain-specific knowledge. We developed QKAR models for two drug toxicity endpoints, drug-induced liver injury (DILI) and drug-induced cardiotoxicity (DICT), using three different knowledge representations with varying levels of knowledge. The representations based on comprehensive knowledge of the drugs yielded better prediction than those with simpler knowledge. Five machine learning algorithms of distinct complexity were applied in QKAR models, and we observed little association between model complexity and performance. Further, we evaluated QKARs against QSARs on the same endpoints using identical datasets. We found that QKARs consistently outperformed QSARs for DILI and DICT. Notably, QKARs demonstrated better capability than QSARs in differentiating drugs with similar structures but different liver toxicity profiles. We also investigated integrating knowledge-based and structure-based representations, Q(K + S)ARs, for further enhanced prediction accuracy. Our findings demonstrate the potential of QKARs as a robust alternative to QSARs, offering additional opportunities in drug toxicity assessments by leveraging both domain-specific knowledge and structural data.

Keywords: computational toxicology; Quantitative Knowledge–Activity Relationships (QKARs); drug-induced cardiotoxicity; drug-induced liver injury; Quantitative Structure–Activity Relationships (QSARs); New Approach Methodologies (NAMs)

Computational toxicology focuses on using in silico approaches to predict the toxic effects of chemicals on living organisms; it has played an increasing role in toxicological research and drug safety studies (Kavlock et al. 2008; Muster et al. 2008; Li et al. 2021, 2023; Sinha et al. 2023; Haßmann et al. 2024; Li et al. 2024; Qu et al. 2025), particularly with the advancement in artificial intelligence (AI) and machine learning (ML). The use of computational approaches is also encouraged by regulatory agencies to support safety assessment. For example, in April 2025, the US Food and Drug Administration (FDA) released the FDA’s *Roadmap to Reducing Animal Testing in Preclinical Safety Studies* https://www.fda.gov/files/newsroom/published/roadmap_to_reducing_animal_testing_in_preclinical_safety_studies.pdf, which emphasizes the adoption of New Approach Methodologies (NAMs) in safety evaluation of consumer products, including AI models. This roadmap is further supported by the FDA Modernization Act 3.0

<https://www.congress.gov/bill/119th-congress/senate-bill/355>, indicating FDA’s broader commitment to modernizing toxicological assessments and reducing reliance on animal use.

Since its inception, computational toxicology has been dominated by Quantitative Structure–Activity Relationship (QSAR) approaches, which are based on the principle that a chemical’s structure determines its biological or toxicological activity (Tropsha et al. 2024). Accordingly, QSAR models widely employ ML to predict toxicity from chemical structures represented by chemical descriptors (Popova et al. 2018; Muratov et al. 2020; Li et al. 2021; Pérez Santín et al. 2021; Chen et al. 2023; Moret et al. 2023). Although QSARs have shown considerable success in toxicology applications (Nantasenamat et al. 2010), their exclusive reliance on chemical structures has limited the scope of their application. This shortcoming is specifically amplified in drug toxicity assessment, where minor structural modifications may

导致毒性发生显著变化。例如，布洛芬和布洛芬酸仅相差一个甲基，但布洛芬通常被认为是一种安全药物（Rainsford 2009），而布洛芬酸因其严重的肝毒性而被从市场上撤回（Goldkind 和 Laine 2006）。许多这样的“药物对”案例凸显了需要更具信息性的化学表示方法，以有效捕捉与生物活性相关的特征。

人工智能（AI）和机器学习（ML）的进步为增强计算毒理学提供了关键能力。AI中最重要的进展之一是生成式人工智能（GenAI）的兴起，它不仅展示了理解文本文件的出色能力，还能够处理其他类型的数据（Cao 等，2023；Fui-Hoon Nah 等，2023）。一个突出的例子是大型语言模型（LLMs），如 ChatGPT，它已展示出总结复杂文件的能力（Liu 等，2024；Ying 等，2024）、通过医学考试的能力（Mbakwe 等，2023；Wang 等，2023）以及在庞大且多样的知识库中推理的能力（Bang 等，2023）。此外，ChatGPT 已被应用于各种拥有丰富上下文信息的领域，包括涉及药物和化学物质的领域，其知识涵盖作用机制、代谢途径和非靶向相互作用（Zheng 等，2023；Pradhan 等，2024）。重要的是，这类知识可以转化为数值向量，即所谓的文本嵌入（text embed）。

我们假设这种嵌入的领域知识可以用于增强药物毒性预测。当与QSARs中使用的传统结构数据结合时，这种方法可能使开发更准确、具有机制信息的毒性预测成为可能。因此，我们提出了定量知识–活性关系（QKARs），它使用通过文本嵌入从大型语言模型（LLMs）获得的特定领域知识来预测药物毒性。我们针对两种药物毒性终点——心脏毒性和肝毒性——开发了QKAR模型，使用了不同的知识表示，知识的范围和深度各不相同。然后，我们使用相同的数据集和相同的开发框架评估了QKARs与传统QSARs的预测性能。还对一些复杂程度不同的机器学习算法进行了比较分析。此外，我们探索了整合基于知识和基于结构的表示对预测的影响。

材料与方法

数据集

DILIst (Thakkar 等，2020) 和 DICTrank (Qu 等，2023) 是最大的药物性肝损伤（DILI）和药物性心脏毒性（DICT）注释数据集，前者主要基于 FDA 药物标签文档，后者则完全基于 FDA 药物标签文档。这些数据集被用于开发本研究中的所有模型。由于生物制剂、混合物、有机金属化合物和无机化合物不适合用于 QSAR 建模，因此在模型开发之前，这些类型的化合物从两个数据集中被移除。QKAR 和 QSAR 模型使用相同的数据集开发，以确保预测性能的公平比较。

为了模拟现实世界的应用，数据集根据药物批准年份被划分用于模型训练和测试。训练集由早些年份批准的药物组成，而测试集包括后来批准的药物，从而使在早期药物上开发的模型能够前瞻性地预测随后批准药物的毒性。对于DILI，使用了DILIst数据

根据药物批准年份 1997 年进行了划分，如我们之前的研究所示（Li 等，2021），得到用于训练的 753 种药物（455 种 DILI 阳性，298 种 DILI 阴性）和用于测试的 249 种药物（149 种 DILI 阳性，100 种 DILI 阴性）。类似地，DICTrank 数据按照 2005 年进行划分，这与我们之前关于 DICT 的研究一致（Qu 等，2025），得到训练集 621 种药物（485 种 DICT 阳性，136 种 DICT 阴性）和测试集 303 种药物（193 种阳性，110 种阴性）。

为 QKARs 生成知识表示

QKAR模型开发的知识表示涉及两个步骤。首先，使用GPT-4o生成药物的知识摘要，并采用两个不同的提示，这些提示在指令的具体程度上有所不同。提示如表1所示。接下来，将知识摘要或药物名称进行嵌入，以使用text-embedding-3-large模型生成知识表示，该模型是一种最先进的基于变压器的方法，可将文本描述转换为高维向量表示（向量长度为3072维）。该嵌入模型通过“套娃表示学习”（Matryoshka Representation Learning）技术在多样且广泛的语料库上进行训练；它能够捕捉语义关系、上下文细节和句法结构，使其在分类、聚类和信息检索等任务中表现有效（Kusupati 等，2022）。该模型不仅分析单个词汇，还会检查这些词汇所处的上下文。

模型开发与评估

应用了五种具有不同算法复杂度的机器学习方法（Wu 等，2021）进行模型开发：K 近邻（KNN）、逻辑回归（LR）、支持向量机（SVM）、随机森林（RF）和极端梯度提升（XGBoost）。

表 1. 使用三组不同输入生成的知识表示。^a

表示 type	输入/提示
药品名称 简单毒	[药品名称] (例如：对乙酰氨基酚) “请总结有关 ^a DILI 的关键信息 用100个字介绍[药品名称]。
药理毒理	“请总结与 ^a DILI 相关的信息以供 [药物名称] 用200个字描述。 包括： (1) 药品名称、类别和适应症。 (2) 用法（剂量、途径、持续时间）。 (3) ^a 与肝脏相关的副作用。 (4) 影响 ^a 肝脏的药物–药物相互作用。 (5) ADME，尤其是 ^a 肝脏代谢和毒性代谢产物。 (6) ^a 肝损伤的机制。 (7) ^a 生物标志物（例如 ALT、AST）。 (8) 患者风险因素（例如年龄、基因）。 (9) 警告、病例报告或监管措施（例如，撤回或黑框警告）。 (10) ^a 肝损伤的潜伏期和严重程度。 “也，指示整体 ^a DILI 关注水平（例如：低、中、高），以及它的比较 对类似药物。

^a 为了 DICT 修改了提示，以便与特定端点的知识相关。

result in significant changes in toxicity. For example, ibuprofen and ibufenac differ by only a single methyl group, but ibuprofen is generally considered a safe drug (Rainsford 2009), whereas ibu-fenac was withdrawn from the market due to its severe hepato-toxicity (Goldkind and Laine 2006). Many of these “drug-pair” cases highlight the need for more informative chemical representations that effectively capture features relevant to biological activity.

Advancements in AI and ML have provided essential capabilities to enhance computational toxicology. One of the most significant advancements in AI is the rise of generative AI (GenAI), which has shown remarkable abilities to not only understand text documents but also other types of data (Cao et al. 2023; Fui-Hoon Nah et al. 2023). A prominent example is Large Language Models (LLMs) such as ChatGPT, which has demonstrated the ability to summarize complex documents (Liu et al. 2024; Ying et al. 2024), pass medical exams (Mbakwe et al. 2023; Wang et al. 2023), and reason over vast and diverse knowledge bases (Bang et al. 2023). In addition, ChatGPT has been applied in various domains rich in contextual information, including those involving drugs and chemicals, with knowledge covering mechanisms of action, metabolic pathways, and off-target interactions (Zheng et al. 2023; Pradhan et al. 2024). Importantly, such knowledge can be converted to numerical vectors, known as text embeddings, which can then serve as input features for various downstream modeling tasks.

We hypothesized that this embedded domain knowledge can be leveraged for enhanced drug toxicity prediction. When combined with traditional structural data used in QSARs, this approach may enable the development of more accurate, mechanistically informed toxicity prediction. Therefore, we proposed Quantitative Knowledge–Activity Relationships (QKARs), which predict drug toxicity using domain-specific knowledge derived from LLMs through text embedding. We developed QKAR models for two drug toxicity endpoints, cardiotoxicity and liver toxicity, using different knowledge representations varying in the scope and depth of knowledge. We then evaluated the predictive performance of QKARs against traditional QSARs using identical datasets and the same development framework. Comparative analyses were also conducted with a range of ML algorithms with varying levels of complexity. Furthermore, we explored the impact of integrating knowledge-based and structure-based representations on prediction performance.

Materials and methods

Datasets

DILIst (Thakkar et al. 2020) and DICTrank (Qu et al. 2023) were the largest datasets with annotation for drug-induced liver injury (DILI) and drug-induced cardiotoxicity (DICT) primarily or completely based on FDA drug labeling documents, respectively. These datasets were used to develop all the models in this study. As biologics, mixtures, organometallics, and inorganics are unsuitable for QSAR modeling, they were removed from both datasets prior to model development. QKAR and QSAR models were developed using identical datasets to ensure a fair comparison in prediction performance.

The datasets were split for model training and testing based on drug approval years to simulate the real-world application. The training set consisted of drugs approved in earlier years, whereas the test set included those approved later, enabling models developed on earlier drugs to prospectively predict the toxicity of subsequently approved drugs. For DILI, the DILIst data

were split based on the drug approval year of 1997, as shown in our previous study (Li et al. 2021), yielding 753 drugs (455 positive and 298 negative for DILI) for training and 249 drugs (149 positive and 100 negative for DILI) for testing. Similarly, the DICTrank data were split at the year 2005, in accordance with our previous study on DICT (Qu et al. 2025), which resulted in a training set of 621 drugs (485 positive and 136 negative for DICT) and a test set of 303 drugs (193 positive and 110 negative).

Generation of knowledge representation for QKARs

The knowledge representation for QKAR model development involved two steps. First, the knowledge summary for a drug was generated using GPT-4o with two different prompts exhibiting varying levels of specificity in instructions. The prompts are shown in Table 1. Next, the knowledge summary or the name of a drug was embedded to generate the knowledge representation using the text-embedding-3-large model, which is a state-of-the-art transformer-based method to convert textual descriptions into high-dimensional vector representations (with a vector length of 3072 dimensions). This embedding model was trained with a technique “Matryoshka Representation Learning” on a diverse and extensive corpus; it captures semantic relationships, contextual details, and syntactic structures, making it effective for tasks such as classification, clustering, and information retrieval (Kusupati et al. 2022). The model not only analyzes individual words but also examines the context in which those words are used, resulting in more accurate and meaningful vector representations. As a result, three knowledge representations (named DrugName, SimpleTox, and PharmTox, respectively) were generated and used for subsequent model development.

Model development and evaluation

Five ML methods with distinct algorithmic complexity (Wu et al. 2021) were applied for model development: K-Nearest Neighbors (KNN), Logistic Regression (LR), Support Vector Machine (SVM), Random Forest (RF), and Extreme Gradient Boosting (XGBoost).

Table 1. Knowledge representations generated using three different sets of input. ^a	
Representation type	Input/prompt
DrugName	[Drug Name] (e.g. acetaminophen)
SimpleTox	“Please summarize key information about ^a DILI for [Drug Name] in 100 words.”
PharmTox	“Please summarize ^a DILI-related information for [Drug Name] in 200 words. “ “Include: (1) drug name, class, and indication. (2) usage (dose, route, duration). (3) ^a liver-related side effects. (4) drug–drug interactions affecting ^a liver. (5) ADME, especially ^a liver metabolism and toxic metabolites. (6) mechanisms of ^a liver injury. (7) ^a biomarkers (e.g. ALT, AST). (8) patient risk factors (e.g. age, genetics). (9) warnings, case reports, or regulatory actions (e.g. withdrawal or black box warnings). (10) latency and severity of ^a liver damage. “Also, indicate the **overall ^a DILI concern level** (e.g. low, moderate, high), and how it compares to similar drugs.”

^a Prompts were modified for DICT where endpoint-specific knowledge is relevant.

这些方法的详细描述可以在文献中找到 (Li 等, 2021)。

对于每种知识表示和机器学习方法的组合, 训练集被以分层方式随机划分 (保持阳性/阴性的比例不变), 用于分配80%的数据进行模型开发, 20%的数据用于内部评估。该过程重复进行100次, 以确保性能估计的稳健性。网络搜索被用来识别每种方法和知识表示组合的最佳超参数。来自100次迭代中20%划分的数据的评估结果被汇总, 以提供对每种知识表示的全面评估。随后, 使用整个训练数据集构建最终的 QKAR模型, 并采用优化后的超参数对测试集进行两种毒性终点的预测。

鉴于两个端点中正负样本分布不平衡, 使用Matthew相关系数 (MCC) 作为主要评估指标。MCC是二分类问题中一个可靠的指标, 能够对模型性能进行平衡评估。其取值范围为−1 到1, 分数越高 (越接近1) 表示对正类和负类的预测越准确。MCC分数越高, 模型性能越好。除了MCC之外, 我们还使用准确率、平衡准确率、敏感性、特异性、F1分数以及AUC-ROC对模型进行评估, 以提供更全面的性能评估。

QKARs与QSARs的比较

QSAR模型针对相同的两个毒性终点 (DILI 和 DICT) 开发, 遵循与 QKAR相同的框架 (即相同的数据集和机器学习方法), 但使用化学描述符代替知识表示。化学描述符使用Mordred (Moriwaki等, 2018) 生成, 这是一种开源工具, 可计算超过1800种二维和三维分子描述符。这些描述符捕捉了物理化学、拓扑和几何性质, 并广泛用于QSAR研究。Mordred描述符在各种性质预测任务中表现出色 (Saini 和 Ramanathan 2022; Tian 等, 2022)。通过保持相同的机器学习流程和评估程序, 该设置允许直接比较这两种方法, 并评估知识表示相对于基于结构描述符的附加价值。

药物组合分析

从我们的最新研究 (Shrimali 等, 2025) 中选择了 DILI 数据集中八对化学结构相似但 DILI 风险不同的药物。从训练集中移除了这八对药物, 而 DILI 数据集中剩余的 986 种药物则用于使用 LR 构建 QKAR 和 QSAR 模型。随后对这八对药物的每种药物进行了 DILI 预测。通过比较 QSAR 模型和 QKAR 模型的能力, 评估它们在区分每对药物中的两种药物方面的效果, 方法是检查引起 DILI 的高毒性药物相对于低毒性药物的相对风险。

基于知识和结构的表示的整合

通过将知识表示与分子描述符连接起来创建了组合特征向量, 使模型能够同时捕捉语义信息和

化合物的理化性质。模型开发和评估遵循与QKAR和QSAR模型相同的流程。

QKAR 模型的结果开发

我们为两个与人体相关的药物毒性终点开发了QKAR模型, 即DILI (Thakkar 等, 2020) 和DICT (Qu 等, 2023), 因为它们是药物失败的前两大原因 (Onakpoya 等, 2016), 并且是现有方法难以解决的安全性问题。我们使用了DILIst (Thakkar 等, 2020) 和DICTrank (Qu 等, 2023) 数据集, 并根据药物批准年份划分数据用于模型训练和测试, 以模拟现实世界的应用。因此, 这些模型在较早批准的药物上进行训练, 并用于前瞻性预测后期批准药物的毒性。

QKAR建模的核心组成部分是知识表示的生成。如图1所示, 知识表示的生成首先是使用GPT-4o总结药物相关的毒性信息。DILI的提示见表1, 而DICT的提示列在表S1中。然后, 将知识摘要或药物名称进行嵌入, 以使用text-embedding-3-large模型生成知识表示。知识嵌入的输入在长度和具体性上有所不同。具体而言, 第一个表示仅基于药物名称。“SimpleTox”表示使用简单提示来查询感兴趣药物的毒性信息。最后, “PharmTox”表示基于药物的药理学和毒理学数据的综合总结。随后, 我们将这三种知识表示分别应用于五种具有不同算法复杂度的机器学习方法 (即KNN、LR、RF、SVM和XGBoost)。

QKAR 模型在嵌入了全面毒性信息的知识表示中表现良好

我们使用一组指标评估了QKAR模型在预测DILI和DICT方面的性能, 包括MCC、准确率、特异性、敏感性、AUC和F1分数 (表S2)。鉴于两个端点中正负样本分布不平衡, MCC被用作主要评估指标。基于不同知识表示和机器学习方法的MCC性能分布通过小提琴图 (用于交叉验证) 和柱状图 (用于测试) 进行了可视化 (图2)。

在DILI预测中, SimpleTox和PharmTox知识表示在所有五种算法中, 在交叉验证和测试中, 一直比DrugName表示产生更好的预测结果, 唯一的例外是在测试结果中的KNN (图2A)。值得注意的是, PharmTox表示相比另外两种表现出更一致的性能, 对不同机器学习方法的敏感性较低 (即变化较小)。

对于DICT预测, 我们在交叉验证中观察到了类似的模式, SimpleTox和PharmTox在QKAR模型中的表现优于DrugName (图2B, 上部)。在测试结果中, 使用PharmTox表示的QKAR模型总体上显示了最佳的预测结果, KNN模型除外。与DrugName表示相比, LR模型在采用SimpleTox和PharmTox知识表示时表现出最显著的改进 (图2B, 下部)。

Detailed descriptions of these methods are available in the literature (Li et al. 2021).

For each combination of knowledge representation and ML method, the training set was randomly split in a stratified manner (the prevalence of positive/negative was maintained) to allocate 80% of the data for model development and 20% for internal evaluation. This process was repeated over 100 iterations to ensure robust performance estimation. A grid search was employed to identify the optimal hyperparameters for each combination of method and knowledge representation. The evaluation results from the 20% splits across the 100 iterations were summarized to provide a comprehensive assessment of each knowledge representation. Thereafter, the entire training dataset was used to build the final QKAR models with the optimized hyperparameters and to make predictions on the test set for two toxicity endpoints, respectively.

Given the imbalanced distribution of positives and negatives in both endpoints, the Matthews Correlation Coefficient (MCC) was used as the primary evaluation metric. MCC is a reliable metric for binary classification, offering a balanced assessment of model performance of a model. Its values range from −1 to 1, with a higher score (closer to 1) indicating accurate predictions for both positive and negative classes. Higher MCC scores indicate better model performance. In addition to MCC, we also evaluated models using accuracy, balanced accuracy, sensitivity, specificity, F1-score, and AUC-ROC to provide a more comprehensive performance assessment.

Comparison of QKARs and QSARs

QSAR models were developed for the same two toxicity endpoints (DILI and DICT) following the same framework as QKARs (i.e. identical datasets and ML methods) with chemical descriptors instead of knowledge representations. The chemical descriptors were generated using Mordred (Moriwaki et al. 2018), an open-source tool that calculates over 1,800 two-dimensional and three-dimensional molecular descriptors. These descriptors capture physicochemical, topological, and geometrical properties and are widely used in QSAR studies. Mordred descriptors have demonstrated strong performance in various property prediction tasks (Saini and Ramanathan 2022; Tian et al. 2022). By maintaining the same ML pipeline and evaluation procedures, this setup allows for a direct comparison between two approaches as well as an assessment of the added value of knowledge representations over structure-based descriptors.

Drug pair analysis

Eight pairs of drugs with similar chemical structures but differing DILI risk were selected from the DILI dataset based on a recent publication from our group (Shrimali et al. 2025). These eight pairs of drugs were removed from the training set, while the remaining 986 drugs in the DILI dataset were used to develop QKAR and QSAR models using LR. DILI predictions were then made for each drug of these eight drug pairs. The QSAR model and the QKAR model were compared for their ability to distinguish the two drugs within a pair by examining the relative risk of the more toxic drug over the less toxic one in causing DILI.

Integration of knowledge- and structure-based representations

The combined feature vectors were created by concatenating knowledge representation with molecular descriptors, allowing the models to capture both semantic insights and

physicochemical properties of compounds. Model development and evaluation followed the same pipeline as the QKAR and QSAR models.

Results Development of QKAR models

We developed QKAR models for two human-relevant drug toxicity endpoints, DILI (Thakkar et al. 2020) and DICT (Qu et al. 2023), as they are the top two causes for drug failure (Onakpoya et al. 2016) and challenging safety concerns to address with existing approaches. We used the DILIst (Thakkar et al. 2020) and DICTrank (Qu et al. 2023) datasets and split the data based on the year of drug approval for model training and testing to simulate the real-world application. Thus, the models were trained on drugs approved in earlier years and tested to prospectively predict the toxicity of those approved later.

The essential component of QKAR modeling is the generation of knowledge representations. As shown in Fig. 1, knowledge representations were generated by first summarizing drug-related toxicity information using GPT-4o. The prompts for DILI are provided in Table 1, whereas those for DICT are listed in Table S1. The knowledge summaries, or the drug names, were then embedded to generate knowledge representations using the text-embedding-3-large model. The input for knowledge embedding varied in length and specificity. Specifically, the first representation was based solely on drug names. The “SimpleTox” representation used a simple prompt to inquire toxicity information about a drug of interest. Lastly, the “PharmTox” representation was based on a comprehensive summary of pharmacological and toxicological data about a drug. We then applied each of the three knowledge representations to five ML methods with different algorithmic complexity (i.e. KNN, LR, RF, SVM, and XGBoost) to develop a total of 15 QKAR models.

QKAR models performed well with knowledge representations embedding comprehensive toxicity information

We evaluated the performance of the QKAR models in predicting DILI and DICT using a set of metrics, including MCC, accuracy, specificity, sensitivity, AUC, and F1 score (Table S2). Given the imbalanced distribution of positives and negatives in both endpoints, MCC was used as the primary evaluation metric. Performance distributions based on MCC across different knowledge representations, and ML methods were visualized using violin plots for cross-validation and bar plots for testing (Fig. 2).

For DILI prediction, the SimpleTox and PharmTox knowledge representations consistently yielded better prediction results than the DrugName representation across all five algorithms in both cross-validation and testing, with the only exception of KNN in test results (Fig. 2A). Notably, the PharmTox representation demonstrated more consistent performance than the other two, exhibiting low sensitivity (i.e. little variation) to different ML methods.

For DICT prediction, we observed similar patterns in cross-validation, with SimpleTox and PharmTox performing better than DrugName in QKAR models (Fig. 2B, top). In the test results, the QKAR models using the PharmTox representation overall showed the best prediction results with the exception of the KNN-based model. LR models showed the most prominent improvement with the SimpleTox and PharmTox knowledge representations compared with the DrugName representation (Fig. 2B, bottom).

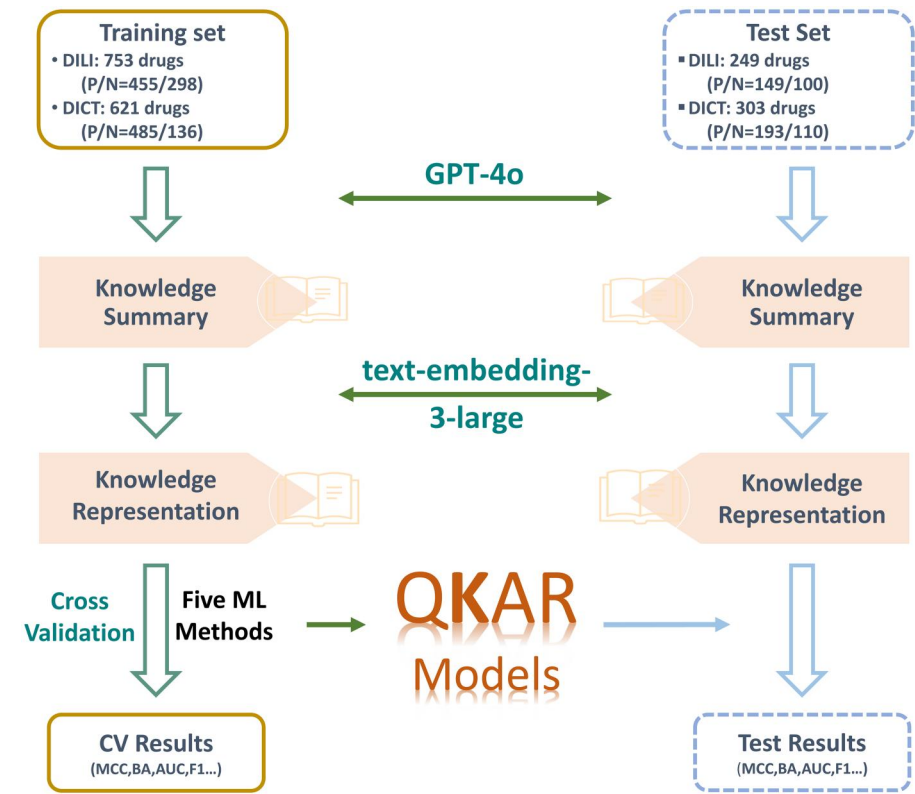


图1. QKAR 开发工作流程。QKAR 模型针对 DILI（药物性肝损伤）和 DICT（药物性心脏毒性）进行了开发。数据集根据药物获批年份划分为训练集和测试集，以模拟真实世界应用。对于每种药物，使用 GPT-4o 和不同提示生成了知识总结。然后，使用 text-embedding-3-large 模型对知识总结进行嵌入，以生成知识表示。应用五种具有不同复杂度的机器学习算法开发 QKAR 模型，并用于对各自的测试集预测 DILI 和 DICT。

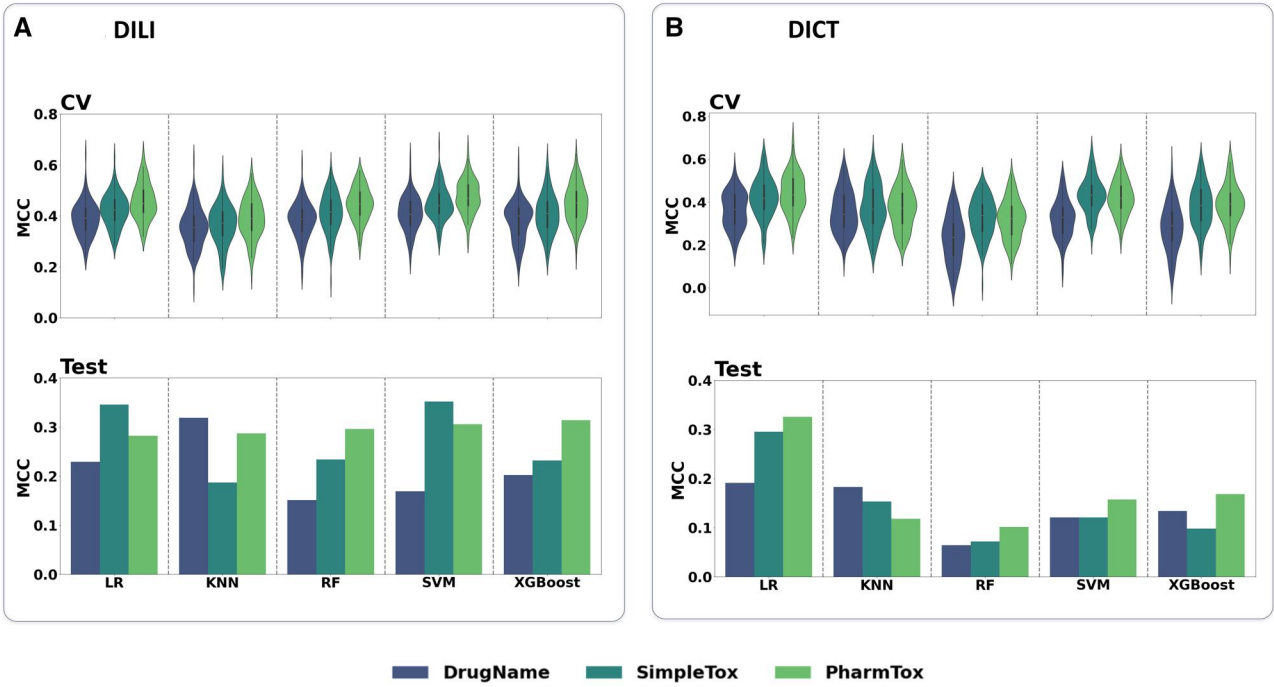


图2. QKAR模型在3种知识表示和5种机器学习方法下的性能。对于DILI（药物性肝损伤）A)和DICT（药物性心脏毒性）B)，小提琴图显示了训练集的交叉验证结果分布，而柱状图显示了测试集的预测结果。CV，交叉验证结果；Test，测试结果。

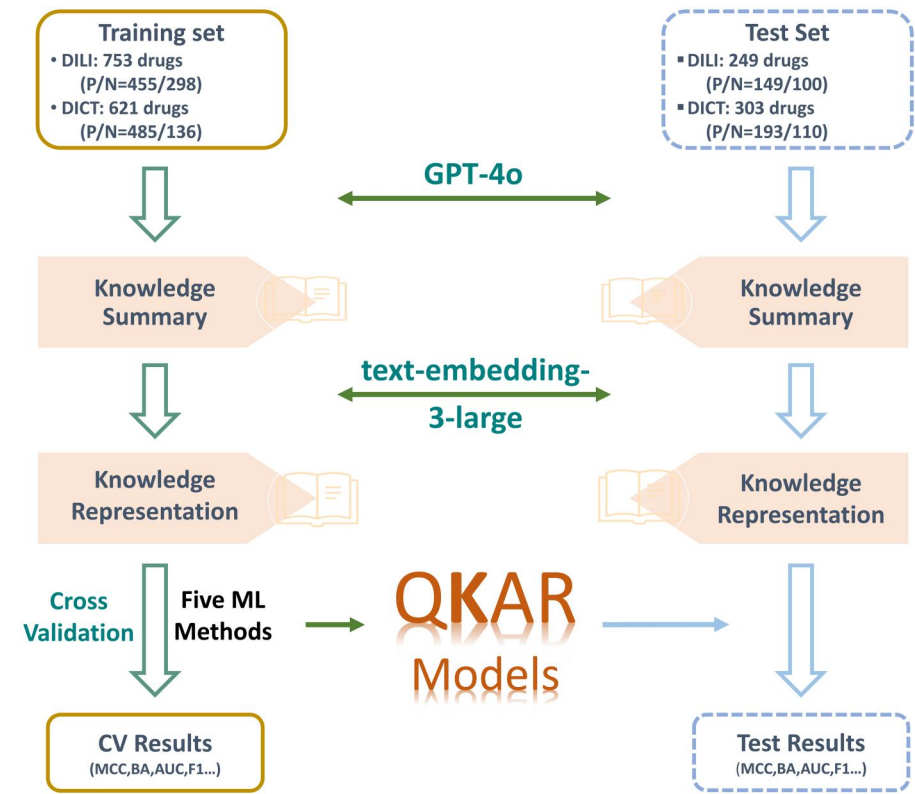


Fig. 1. Workflow for QKAR development. QKAR models were developed for both DILI (Drug-Induced Liver Injury) and DICT (Drug-Induced Cardiotoxicity). The datasets were split for training and testing based on drug approval years to simulate real-world applications. For each drug, a knowledge summary was generated using GPT-4o with different prompts. The knowledge summary was then embedded using the text-embedding-3-large model to generate the knowledge representation. Five ML algorithms with varying levels of complexity were applied to develop QKAR models, which were then used to predict DILI and DICT on the respective test sets.

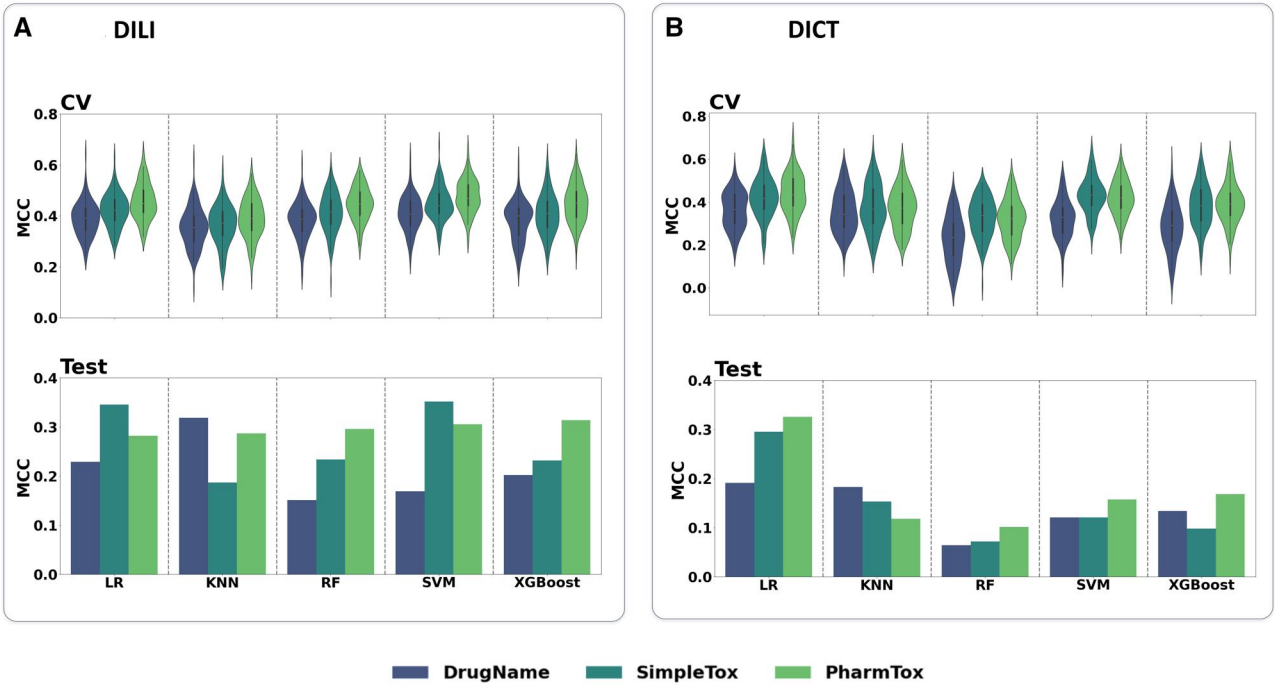


Fig. 2. Performance of the QKAR models across 3 knowledge representations and 5 ML methods. For both DILI (Drug-Induced Liver Injury) A) and DICT (Drug-Induced Cardiotoxicity) B), the violin plots show the distributions of cross-validation results on the training set, whereas the bar charts display the prediction results on the test sets. CV, cross-validation results; Test, test results.

这些结果表明, 相较于较简单的输入 (即仅药物名称), 全面的毒性信息使QKARs的预测性能更高, 这突显了领域特定知识在准确药物毒性评估中的重要性。此外, 对于测试集, 较复杂的机器学习方法在DILI或DICT的预测准确性上并未表现出更好的效果。相反, 作为简单模型之一的LR在DICT的预测上远优于其他四种模型。结果显示, 机器学习方法的复杂性与预测性能之间几乎没有关联, 这表明算法的选择可能并非关键因素。

QKARs 在 DILI 和 DICT 预测中均优于 QSARs

为了评估QKAR相对于传统QSAR的性能, 我们使用与QKAR开发相同的数据集和模型开发流程开发了QSAR模型; 唯一的区别是QSAR使用基于结构的描述符, 而QKAR使用知识表示。对于DILI和DICT, 无论使用哪种机器学习方法, QKAR模型在交叉验证中始终优于QSAR模型, 如小提琴图所示 (图3, 上部)。逻辑回归 (LR) 被证明是一种简单而有效的方法, 表现优异; 因此, 我们进一步检查了交叉验证过程中使用LR生成的各个模型。由于我们进行了5折交叉验证, 共迭代20次, 因此总共生成了100个不同的训练-测试拆分。对于每个拆分, 使用LR开发了两个模型: 一个采用QKAR方法, 另一个采用QSAR方法。比较分析显示, QKAR在所有100个DILI模型中以及在其中的90个模型中始终优于QSAR。

结果表明, QKAR 可能作为 QSAR 的一种稳健替代方法, 用于增强药物毒性预测。

QKARs 在区分具有大肝损伤风险差异的结构相似药物方面表现出比 QSARs 更高的能力

基于化学结构的方法, 例如QSARs, 往往难以区分具有相反毒性的结构相似化合物 (Maggiore 2006)。为了评估QKARs在这些情况下的预测分辨率, 我们使用LR方法比较了QKAR模型和QSAR模型在八对药物上的表现; 每对药物均由两种结构相似但肝毒性差异较大的药物组成 (图4)。我们通过将模型预测的较高毒性药物的DILI概率除以较低毒性药物的DILI概率, 来计算药物对在引起DILI方面的相对风险。相对风险大于1表示模型能够区分具有不同引起DILI潜力的两种药物。相对风险越大, 模型区分药物对的能力越强。基于这一初步分析, 我们观察到QKAR模型在所有八对药物中均实现了毒性比大于1, 而QSAR模型

整合基于知识和结构的表示以增强毒性预测

我们还研究了将药物的知识和结构整合是否能够进一步提高毒性预测。该方法被称为 Q(K+S)ARs (定量知识与结构-活性关系), 旨在利用生物医学背景的互补优势和

These results suggested that the comprehensive toxicity information enabled higher predictive performance in QKARs than the simpler input (i.e. drug name only), highlighting the importance of domain-specific knowledge for accurate drug toxicity evaluation. Furthermore, the more complex ML methods did not show better prediction accuracy for either DILI or DICT with the test sets. On the contrary, LR, as one of the simple models, achieved much better prediction for DICT than the other four models. The results showed little association between the complexity of the ML methods and the prediction performance, indicating that the selection of algorithms may not be critical.

QKARs outperformed QSARs in both DILI and DICT prediction

To evaluate the performance of QKARs relative to the conventional QSARs, we developed QSAR models using the identical datasets and model development pipeline as for QKAR development; the only difference was to use structure-based descriptors for QSARs instead of knowledge representations for QKARs. For both DILI and DICT, QKAR models consistently outperformed QSAR models in cross-validation regardless of the ML methods that were used, as shown in the violin plots (Fig. 3, top). LR demonstrated to be a simple yet effective method with strong performance; therefore, we further examined the individual models generated during cross-validation with LR. As we performed 5-fold cross-validation with 20 iterations, a total of 100 distinct train-test splits were generated. For each split, two models were developed with LR: One using the QKAR approach and the other using QSAR. Comparative analysis showed that QKAR consistently outperformed QSAR in all 100 DILI models and in 90 out of 100 DICT models (Fig. S1). Furthermore, we compared the model performance on the test set and observed that QKARs again showed superior performance to QSARs (Fig. 3, bottom). Our

results suggested that QKAR may serve as a robust alternative to QSAR for enhanced drug toxicity prediction.

QKARs exhibited higher capability than QSARs in differentiating structurally similar drugs with large differences in DILI risks

Chemical structure-based approaches, such as QSARs, tend to have difficulties separating structurally similar compounds with opposite toxicities (Maggiore 2006). To assess the predictive resolution of QKARs on these cases, we compared the performance of the QKAR and QSAR model using the LR method for eight drug pairs; each pair consisted of two structurally similar drugs but with large differences in hepatotoxicity (Fig. 4). We calculated the relative risk of the two drugs in a pair in causing DILI by dividing the model-predicted DILI probability of the more toxic drug by that of the less toxic one. A relative risk greater than 1 indicates the ability of a model to distinguish between two drugs with different potential to cause DILI. The greater the relative risk was, the more capable a model was of differentiating the drug pair. Based on this preliminary analysis, we observed that the QKAR model achieved a toxicity ratio greater than 1 for all eight drug pairs, whereas the QSAR model surpassed this threshold for only five pairs. Our results showed that the QKAR model appeared to exhibit better capability than the QSAR model in differentiating structually-similar drugs with varying toxicity levels.

Integrating knowledge- and structure-based representations for enhanced toxicity prediction

We also investigated whether integrating the knowledge and structure of drugs would further improve toxicity prediction. This approach, referred to as Q(K+S)ARs (Quatitative Knowledge&Structure-Activity Relationships), was designed to leverage the complementary strengths of biomedical context and

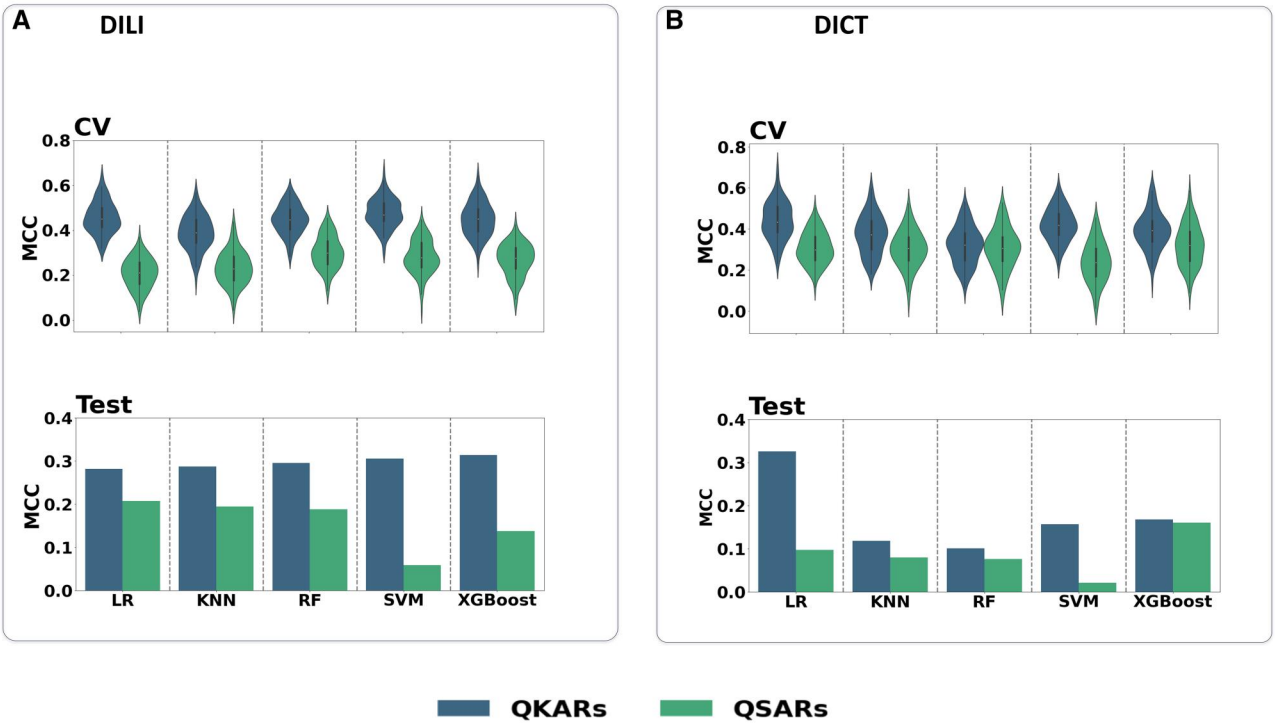


图3. QKAR与QSAR模型的性能比较。对于DILI (药物性肝损伤) A) 和DICT (药物性心脏毒性) B), 小提琴图显示了训练集交叉验证结果的分布, 柱状图显示了测试集的预测结果。QKAR的结果基于PharmTox表示。

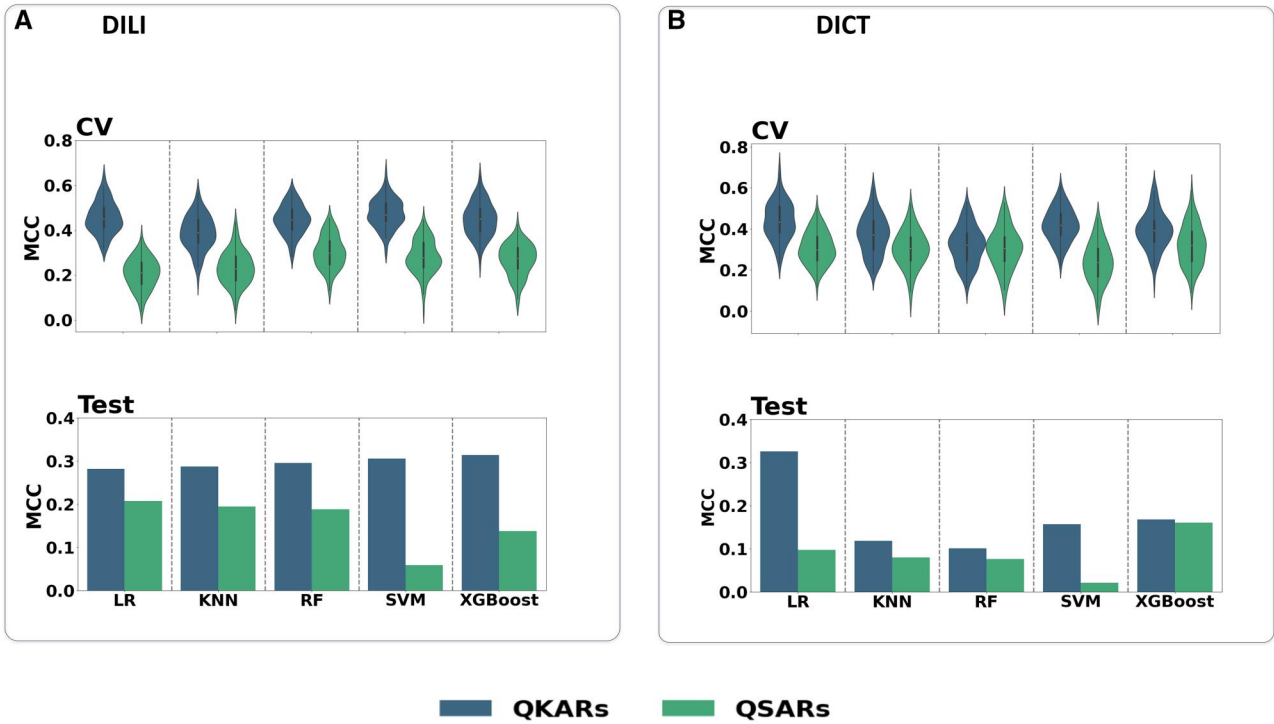


Fig. 3. Performance comparison between QKAR and QSAR models. For both DILI (Drug-Induced Liver Injury) A) and DICT (Drug-Induced Cardiotoxicity) B), the violin plots show the distributions of cross-validation results on the training sets, and the bar chart display the prediction results on the test sets. The results from QKARs were based on the PharmTox representations.

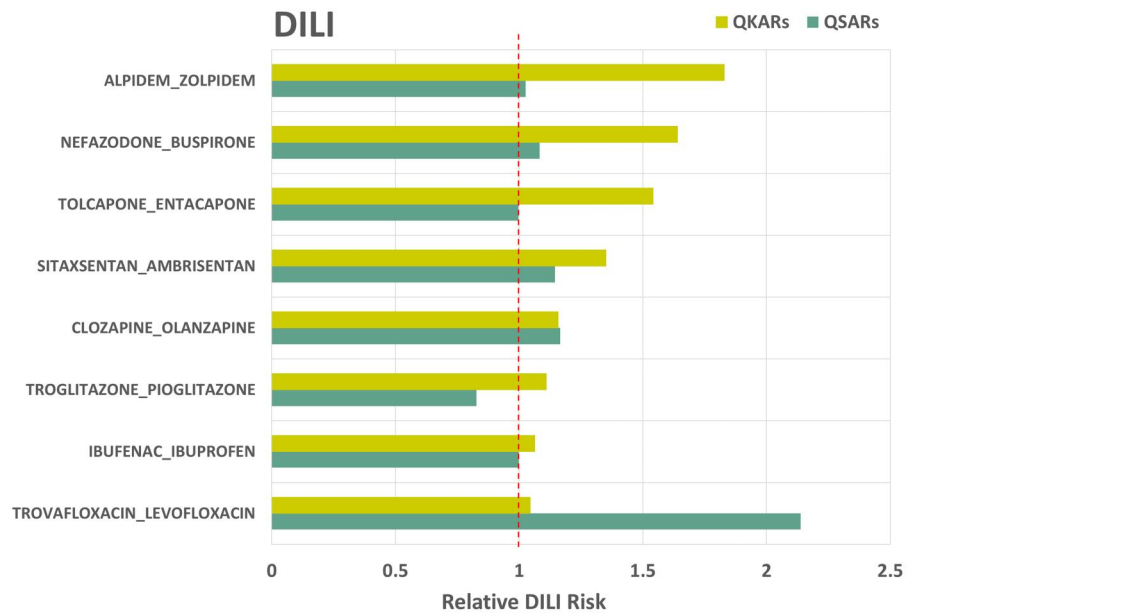


图4. QKAR与QSAR在区分具有相似结构但不同引起DILI潜力的药物中的比较。每根柱表示基于模型预测，两种药物引起DILI的相对风险。相对风险大于1（红色虚线右侧）表明模型能够区分毒性较高的药物和毒性较低的药物。相对风险越大，模型区分药物对的能力越强。QKAR和QSAR模型均采用逻辑回归（LR）方法开发。QKAR模型使用了PharmTox表示法进行开发。

分子描述符。为此，我们通过结合基于知识和基于结构的特征，并遵循图1所示的相同建模流程，开发了Q(K + S)AR模型。这些模型在五种机器学习方法下进行了评估，并与DILI和DICT端点的QSAR和QKAR模型进行了比较。对于DILI（图5A）和DICT（图5B），Q(K + S)AR模型始终优于QSAR模型；然而，相对于QKAR模型的优势则较为有限。

讨论

在本研究中，我们开发了一种新型框架QKARs，通过文本嵌入利用大型语言模型（LLMs）的上下文文化化学知识进行毒性预测，重点关注肝损伤（DILI）和药物诱导肾损伤（DICT）——这是药物开发和安全评估中最具挑战性的两个终点。我们使用相同的数据集和工作流程，对QKARs和QSARs进行了正面对比，采用了五种算法复杂度不同的机器学习算法。QKAR模型在训练集和测试集上始终优于QSAR模型，这表明QKARs在预测毒理学中的潜在应用价值。我们还研究了基于知识和基于结构的表示整合，即Q(K+S)ARs，以进一步提升预测准确性。

QSAR 在预测毒理学中发挥了关键作用，通过模拟化学结构与生物活性之间的关系（Hansch 等，1962；Tropsha 等，2024）。传统和AI增强的结构表示方法包括指纹（例如 MACCS [Durant 等 2002]，ECFP [Rogers 和 Hahn 2010]）、描述符（例如 Mold2 [Hong 等 2008]，以及 Dragon [Mauri 等 2006]）到基于图和 SMILES 的嵌入（Goh 等，2017；Hirohara 等，2018；Chakravarti 和 Alla，2019；Chen 等，2021；Hung 和 Gini，2021），这些方法拓宽了 QSAR 的应用范围以支持多样化的应用。最近的方法包括 MoLFormer（Ross 等，2022）和 ChemBERTa（Chithrananda 等，2020），它们利用大型化学结构数据集和 transformer。

通过 SMILES 表示学习分子特征的架构（Duvenaud 等人 2015；Winter 等人 2019；Chithrananda 等人 2020；Fabian 等人 2020；Irwin 等人 2022；Ross 等人 2022；Xia 等人 2022；Yüksel 等人 2023；Zhou 等人 2023）。然而，QSAR 的基本原理——化学结构决定生物活性——具有固有的局限性，因为生物活性受多种因素影响，而不仅仅是结构，尤其是在药物毒性预测的背景下。大型语言模型（LLM）的出现提供了一种基于知识的替代方案，通过从大量科学文献中提取化学和生物背景信息，从而替代传统 QSAR 的分子表示（Biswas 2023；Guo 等人 2023；Zheng 等人 2023；Pradhan 等人 2024）。通过生成捕捉这些科学数据的基于文本的嵌入，可以使用类似于传统 QSAR 描述符的知识衍生描述符来补充甚至增强毒性预测。

QKAR相对于QSAR在潜在应用方面的广泛范围总结如表2所示。QSAR主要局限于小分子有机化合物，因此在处理复杂混合物、产品配方以及区分具有显著生物活性差异的结构相似化合物方面存在困难（Maggiora 2006；Weaver 和 Gleeson 2008；Fourches 等，2010；Cherkasov 等，2014）。它们高度依赖基于结构的特征，并且需要化学预处理和特征选择，这可能导致信息丢失。相比之下，QKAR能够对包括生物制剂、大分子、纳米材料、无机物和混合物在内的各种化学产品建模，因为它们使用知识表示而不是化学结构。QKAR还能够捕捉配方差异，避免依赖结构相似性，能够整合多领域数据（化学、生物、环境、人类和动物），并且无需进行特征选择和化学预处理。

尽管QKARs具有潜在的前景，但要充分实现其能力，必须解决若干挑战。首先，QKARs依赖于特定领域的知识；因此，进一步

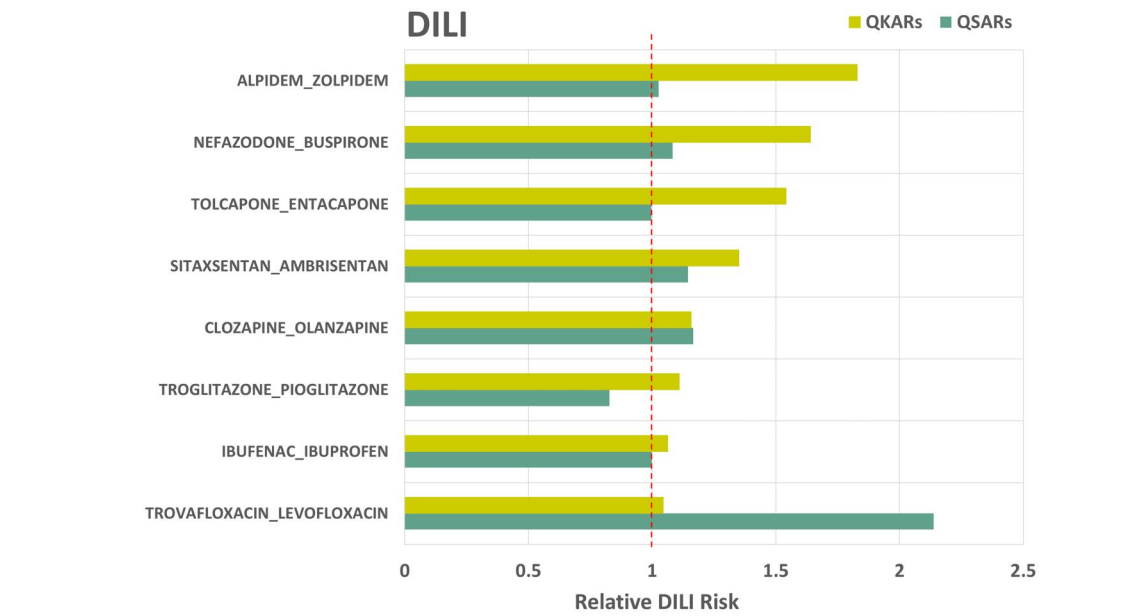


Fig. 4. Comparison of QKARs and QSARs in differentiating drugs with similar structure but different potential to cause DILI. Each bar represents the relative risk of two drugs in causing DILI based on a model’s prediction. A relative risk greater than 1 (to the right of the red dashed line) indicated the ability of a model in differentiating the more toxic drug from the less toxic one. The greater the relative risk was the better capability the models had in distinguishing a drug pair. Both QKAR and QSAR models were developed using the logistic regression (LR) method. The PharmTox representation was used to develop the QKAR model.

molecular descriptors. For that, we developed the Q(K + S)AR models by combining the knowledge-based and structure-based features and following the same modeling pipeline as depicted in Fig. 1. These models were evaluated across five ML methods and compared against both QSARs and QKARs for both the DILI and DICT endpoints. For both DILI (Fig. 5A) and DICT (Fig. 5B), the Q(K + S)AR models consistently outperformed QSAR models; however, their advantage over QKAR models was modest.

Discussion

In this study, we developed a novel framework, QKARs, that leverages contextual chemical knowledge from LLMs via text embeddings for toxicity prediction, focusing on DILI and DICT—two of the most challenging endpoints in drug development and safety assessment. We conducted a head-to-head comparison between QKARs and QSARs using the same datasets and workflow, employing five ML algorithms with varying degrees of algorithmic complexity. QKAR models consistently outperformed QSAR models in both training and test sets, which demonstrated the potential utility of QKARs in predictive toxicology. We also investigated integrating knowledge-based and structure-based representations, Q(K + S)ARs, for further enhanced prediction accuracy.

QSARs have played a pivotal role in predictive toxicology by modeling the relationship between chemical structure and biological activity (Hansch et al. 1962; Tropsha et al. 2024). Traditional and AI-enhanced structure representations range from fingerprints (e.g. MACCS [Durant et al. 2002], ECFP [Rogers and Hahn 2010]), descriptors (e.g. Mold2 [Hong et al. 2008], and Dragon [Mauri et al. 2006]) to graph- and SMILES-based embeddings (Goh et al. 2017; Hirohara et al. 2018; Chakravarti and Alla 2019; Chen et al. 2021; Hung and Gini 2021), which have broadened the QSAR landscape to support diverse applications. Recent approaches, including MoLFormer (Ross et al. 2022) and ChemBERTa (Chithrananda et al. 2020), leverage large chemical structure datasets and transformer

architectures to learn molecular features through SMILES representations (Duvenaud et al. 2015; Winter et al. 2019; Chithrananda et al. 2020; Fabian et al. 2020; Irwin et al. 2022; Ross et al. 2022; Xia et al. 2022; Yüksel et al. 2023; Zhou et al. 2023). However, the fundamental principle of QSARs—that chemical structure solely determines biological activity—has inherent limitations, as biological activity is influenced by multiple factors beyond structure, particularly in the context of drug toxicity prediction. The advent of LLMs offers a knowledge-driven alternative to traditional QSAR molecular representations by extracting chemical and biological context from vast scientific literature (Biswas 2023; Guo et al. 2023; Zheng et al. 2023; Pradhan et al. 2024). By generating text-based embeddings that capture these scientific data, knowledge-derived descriptors analogous to traditional QSAR descriptors can be used to complement or even enhance toxicity prediction beyond the limitations of structure-based models.

The broad scope of potential applications of QKARs over QSARs is summarized in Table 2. QSARs are mainly limited to small organic molecules and thus struggle with complex mixtures, product formulations, and distinguishing structurally similar compounds with large differences in biological activities (Maggiora 2006; Weaver and Gleeson 2008; Fourches et al. 2010; Cherkasov et al. 2014). They rely heavily on structure-based features and require chemical preprocessing and feature selection, which may lose information. In contrast, QKARs can model a wide variety of chemical products, including biologics, macromolecules, nanomaterials, inorganics, and mixtures, because they use knowledge representation rather than chemical structure. QKARs can also capture formulation differences, avoid reliance on structural similarity, potentially integrates multi-domain data (chemical, biological, environmental, human, and animal), and eliminate the need for feature selection and chemical preprocessing.

Despite the promising potential of QKARs, several challenges must be addressed to fully realize their capabilities. First, QKARs rely on domain-specific knowledge; therefore, further

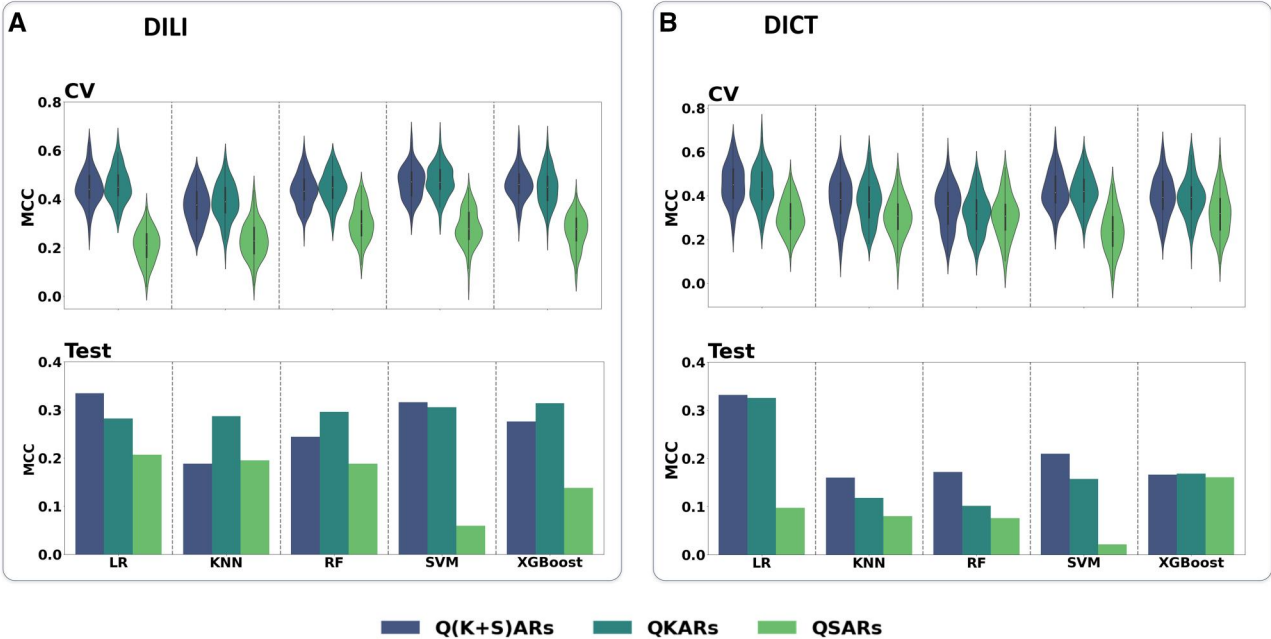


图5. Q(K+S)AR、QKAR 和 QSAR 模型的性能比较。对于 DILI（药物性肝损伤）A）和 DICT（药物性心脏毒性）B），小提琴图显示了训练集交叉验证结果的分布，而柱状图显示了测试集的预测结果。Q(K+S)ARs 和 QKARs 的结果基于 PharmTox 表征。

表2. QKAR在传统QSAR之外的潜在多样化应用。

范围	定量构效关系	QKARs
化学产品	主要用于小型有机分子，在处理他人时构成挑战化学产品的种类。	可以模拟各种广泛的产品，例如作为生物制剂（例如单克隆抗体），大分子，纳米材料，有机金属化合物，无机化合物，草药药物、膳食补充剂等
复杂混合物	表示和建模能力有限混合物，多组分配方，或其他结构复杂的成分。	因为一个复杂的混合物被表示了知识，它可以很容易地被修改
产品配方	药物活动通常与……相关它们的离子形式和盐-对离子未被表示的交互适当地与基于结构的化学-校准描述符，更不用说信息与剂量和给药途径相关。	知识表示可以捕捉到产品配方的差异
活性悬崖	结构上相似的难以区分具有较大差异的化合物生物活性，尤其是在药物毒性中明显	结构相似性在……中作用不大可能使……有所区分的知识生物学差异。
输入特征	仅限于基于结构的化学信息。	专注于知识，允许整合多领域知识（化学，生物的、环境的、动物的，以及人类数据）。
建模复杂性	化学预处理和特征通常需要选择，这可能导致信息丢失。	既没有化学预处理，也没有特征需要选择。

对于将QKARs应用于数据匮乏的化学品（如药物开发早期阶段的候选药物），需要进行调查。其次，不同提示引入的变异可能导致知识总结的结果不同，从而影响基于知识的模型性能。值得注意的是，即使是相同的提示也可能产生不同的响应，这对结果的可重复性和可解释性的一致性提出了挑战。最后，用于监管用途的QKAR模型的生命周期维护必须应对这些挑战。

与不断发展的大型语言模型（LLM）和文本嵌入方法相关，这些都可能影响预测的一致性和可重复性。这表明必须建立定期模型更新、标准化和验证的策略。未来的工作应侧重于开发针对毒性特定的嵌入，并使用精选的数据集（如 FDA 标签文件）对 LLM 进行微调。这些努力将有助于将 QKAR 建立为一个多功能、可扩展的工具，以推进预测毒理学并支持更安全的药物开发。

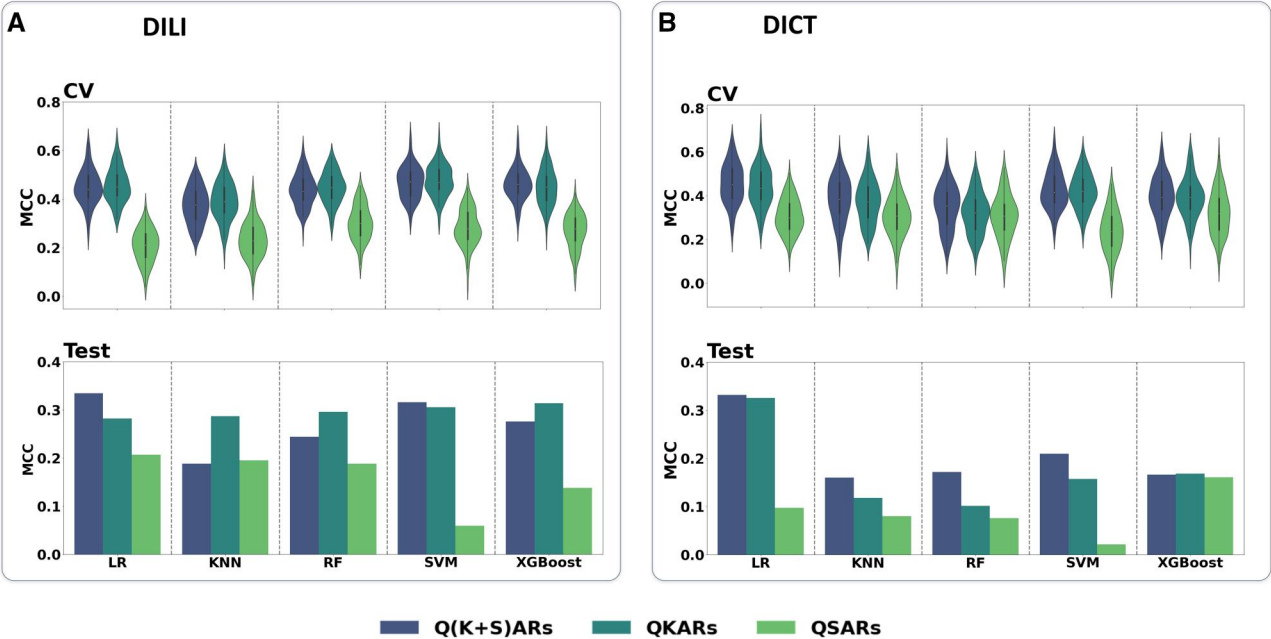


Fig. 5. Performance comparison among Q(K+S)AR, QKAR, and QSAR models. For both DILI (Drug-Induced Liver Injury) A) and DICT (Drug-Induced Cardiotoxicity) B), the violin plots show the distributions of cross-validation results on the training set, whereas the bar charts display the prediction results on the test set. The results from Q(K+S)ARs and QKARs were based on the PharmTox representations.

Table 2. Potential diverse applications of QKARs beyond traditional QSARs.

Scope	QSARs	QKARs
Chemical products	Primarily for small organic molecules, posing a challenge in dealing with other types of chemical products.	Can model a broad range of products, such as biologics (e.g. monoclonal antibodies), macromolecules, nanomaterials, organometallics, inorganics, herbal medicines, dietary supplements, etc.
Complex mixtures	Limited capability to represent and model mixtures, multi-component formulations, or other structurally complex constituents.	Because a complex mixture is represented with knowledge, it can be readily modeled.
Product formulation	Drug activities are often associated with their ionic forms and salt-counterion interactions, which are not represented appropriately with structure-based chemical descriptors, let alone the information relating to dose and administration route.	Knowledge representation can capture the differences in product formulation.
Activity cliffs	Difficult to distinguish structurally similar compounds with large differences in biological activities, which is especially pronounced in drug toxicity.	Structural similarity plays little role in knowledge that may differentiate the biological differences.
Input features	Restricted to structure-based chemical information.	Focused on knowledge, allowing integration of multi-domain knowledge (chemical, biological, environmental, animal, and human data).
Modeling complexity	Chemical preprocessing and feature selection are often required, which may lead to information loss.	Neither chemical preprocessing nor feature selection is needed.

investigation is required for applying QKARs to data-poor chemicals, such as drug candidates in the early stages of drug development. Second, variability introduced by different prompts may lead to different outcomes in knowledge summarization and, consequently, knowledge-based model performance. It is worth noting that even the same prompt may result in different responses, presenting a concern for reproducibility in results and consistency in explainability. Lastly, lifecycle maintenance of QKAR models for regulatory use must address challenges

associated with the evolving nature of LLMs and text embedding methods, all of which may impact prediction consistency and reproducibility. This suggests that strategies for regular model updates, standardization, and validation must be established. Future work should focus on developing toxicity-specific embeddings and fine-tuning LLMs with curated datasets, such as FDA labeling documents. These efforts will help establish QKAR as a versatile, scalable tool for advancing predictive toxicology and supporting safer drug development.

理论上，将化学结构信息与相关领域知识结合应该能够提高预测性能。为了探索这一点，我们尝试了一种整合方法，即 Q(K $\bowtie$ S)ARs，通过将基于知识的描述符与基于结构的化学描述符串联作为模型输入。然而，我们并未观察到比单独使用 QKARs 明显更好的效果。我们推测，这可能是由于这两种描述符类型的本质差异：单个化学描述符独立地表示特定的物理化学性质，而 QKAR 描述符则整体代表综合的领域知识。简单地将两者串联可能会稀释或损害用于建模的化学关键信息。对于在描述符层、模型层或通过混合方法整合化学信息和知识，有多种策略可供探索，这仍然是一个需要进一步研究的领域。

这项研究强调了QKARs通过利用化学和生物学知识进行可靠的安全性评估，从而推动计算毒理学发展的潜力。GenAI和文本嵌入技术的整合为预测毒理学的进一步创新开辟了新的机会。未来的工作应集中于开发在毒理学和药理学数据集上进行微调的领域专用大型语言模型（LLMs），以丰富和增强知识表示。将这些方法与多模态数据源相结合，例如高通量筛选结果、组学数据和临床报告，可能大大提高模型的泛化能力和预测准确性。此外，模型解释技术对于确保基于AI的毒性预测在监管考虑中所需的可解释性和可靠性至关重要。最终，这些创新有可能通过结构为中心的方法和以知识驱动的方法扩展预测毒理学工具箱。

补充材料

补充材料可在《毒理学科学》在线获取。

资金

这项研究的资金由美国食品药品监督管理局提供。

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数据可用性

用于开发和评估 QKAR 的数据集是从 DILiSt (<https://www.fda.gov/science-research/liver-toxicity-knowledge-base-ltkb/drug-induced-liver-injury-severity-and-toxicity-dilist-dataset>) 和 DICTrank (<https://www.fda.gov/science-research/bioinformatics-tools/drug-induced-cardiotoxicity-rank-dictrank-dataset>) 筛选的。基于知识的表示和基于结构的表示已提交到 <https://github.com/TingLi2016/QKAR>。

代码可用性

QKARs 是使用开源 Python（版本 3.7.3）开发的。源代码可在 <https://github.com/TingLi2016/QKAR> 获取。

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In theory, combining chemical structure information with relevant domain knowledge should enhance predictive performance. To explore this, we attempted an integration approach, Q(K+S)ARs, by concatenating knowledge-based descriptors with structure-based chemical descriptors as model inputs. However, we did not observe a significant improvement over QKARs alone. We speculated that this was due to the fundamentally different nature of the two descriptor types: Individual chemical descriptors represent specific physico-chemical properties independently, whereas QKAR descriptors collectively represent integrated domain knowledge. Simply concatenating the two may dilute or compromise key information about the chemicals that is utilized for modeling. There are multiple strategies that can be explored for integrating chemical information and knowledge, at the descriptor level, the model level, or through hybrid approaches, and this remains an area requiring further investigation.

This study highlights the potential of QKARs to advance computational toxicology by leveraging chemical and biological knowledge for reliable safety assessments. The integration of GenAI and text embedding technologies opens new opportunities for further innovation in predictive toxicology. Future efforts should focus on developing domain-specific LLMs fine-tuned on toxicology and pharmacology datasets to enrich and enhance knowledge representations. Combining these approaches with multimodal data sources, such as high-throughput screening results, omics profiles, and clinical reports, may greatly improve model generalizability and predictive accuracy. Moreover, model interpretation techniques will be critical to ensuring the explainability and reliability needed for regulatory consideration of AI-based toxicity predictions. Ultimately, these innovations have the potential to expand the predictive toxicology toolbox with both structure-centered methodologies and knowledge-driven approaches, thereby accelerating safer drug development and personalized risk assessment.

Supplementary material

[Supplementary material](#) is available at *Toxicological Sciences* online.

Funding

Funding for this research was provided by the US Food and Drug Administration.

Conflicts of interest. None declared.

Data availability

The datasets used to develop and evaluate QKARs were curated from DILiSt (<https://www.fda.gov/science-research/liver-toxicity-knowledge-base-ltkb/drug-induced-liver-injury-severity-and-toxicity-dilist-dataset>) and DICTrank (<https://www.fda.gov/science-research/bioinformatics-tools/drug-induced-cardiotoxicity-rank-dictrank-dataset>). The knowledge-based representations and structure-based representations have been deposited at <https://github.com/TingLi2016/QKAR>.

Code availability

QKARs was developed using open-source Python (version 3.7.3). The source code is available at <https://github.com/TingLi2016/QKAR>.

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This article reflects the views of the authors and does not necessarily reflect those of the US Food and Drug Administration. Any mention of commercial products is for clarification only and is not intended as approval, endorsement, or recommendation.

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